MICROSOFT FABRIC / POWER BI

AdventureWorks Analytics Implementation

Complete, evidence-led walkthrough - from raw CSV ingestion and dimensional modeling to a five-page executive dashboard and pipeline orchestration.

04	05	01
Lakehouse tables	Report pages	ETL pipeline

DOCUMENTATION EDITION • 10 NOVEMBER 2026

01 OVERVIEW

Project at a glance

What was implemented and what the analytical solution delivers.

6

CSV source groups

customer, geography, products, hierarchy, sales

4

Analytical Delta tables

DimCustomer, DimProduct, FactInternetSales, DimDate

5

Interactive report pages

Executive to cost analysis

2

Core pipeline activities

Dataflow and Notebook

BUILD CHAIN

CSV files – **Power Query** – **Lakehouse** – **Direct Lake** – **Power BI**

Functional outcomes

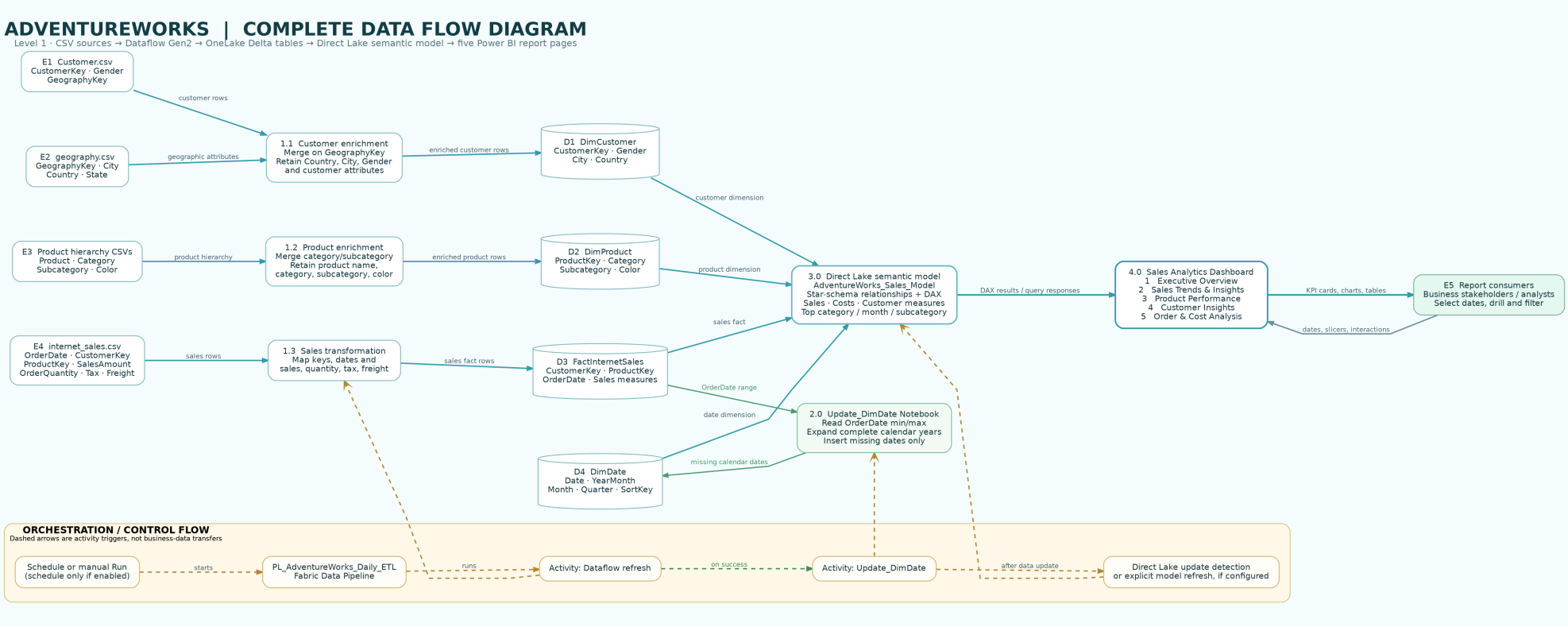
- Consolidated sales, product, customer and geography data into a star-schema reporting model.
- Built reusable DAX metrics, date slicing, KPI trend indicators and five themed report pages.
- Configured a pipeline with a Dataflow Gen2 refresh followed by a date-maintenance notebook.

What this document proves

- Exact Fabric item names and major configuration decisions are documented.
- Screenshots are taken from the report created during the implementation.
- Schedule activation and automatic Direct Lake synchronization require final operational verification.

End-to-end data flow

Data movement is shown with solid connections; orchestration and refresh dependencies are control flows.



Read left to right: input CSVs → Dataflow Gen2 enrichment and typed facts → Lakehouse Delta tables → Direct Lake semantic model → dashboard. A Notebook maintains DimDate from actual OrderDate values. Pipeline activities orchestrate ETL in success order.

Step 01-04 | Prepare source data

The solution starts with source CSV files; no new sales values are synthesized by the pipeline.

01 Create the Fabric workspace

Open Fabric → Workspaces → create/select AdventureWorksAnalytics. Keep the same workspace for dataflow, Lakehouse, semantic model, report and pipeline.

02 Create AdventureWorksLakehouse

Workspaces → New item → Lakehouse. Name it AdventureWorksLakehouse and use it as the destination for analytical Delta tables.

03 Add source CSV connections

Create DF_AdventureWorks_Transformation as Dataflow Gen2; import each source through the CSV/file connectors accessible to your Fabric workspace.

04 Inspect and type columns

Check identifiers (whole number), names and geography (text), OrderDate/ShipDate (date), SalesAmount/TaxAmt/Freight (decimal). Remove invalid rows intentionally.

SOURCE INVENTORY

Input file / group	Important fields	Target / purpose
Customer.csv	CustomerKey, names, gender, GeographyKey	DimCustomer
geography.csv	GeographyKey, City, StateProvince, Country	Customer enrichment
Product.csv	ProductKey, product name, color, hierarchy keys	DimProduct
Subcategory CSV	Subcategory, category relationship	Product enrichment
Category CSV	Category names and category keys	Product enrichment
internet_sales.csv	CustomerKey, ProductKey, OrderDate, amounts	FactInternetSales

Step 05-08 | Design Dataflow Gen2

Transform raw source queries into a lean star-schema-ready analytical layer.

05 Customer + geography merge

In DimCustomer, merge geography by GeographyKey. Expand City, Country and needed region fields; retain CustomerKey and customer attributes.

06 Product hierarchy merge

Merge product records with subcategory and category staging queries. Retain ProductKey, EnglishProductName, Category, Subcategory and Color.

07 Sales fact cleanup

In FactInternetSales, preserve ProductKey, CustomerKey, OrderDate, OrderQuantity, SalesAmount, TaxAmt and Freight. Validate numeric/date types.

08 Configure destinations and run

Map only DimCustomer, DimProduct and FactInternetSales to existing Lakehouse destinations; validate schema and replace behavior before refresh.

Why staging queries stay separate

- Geography, Category and Subcategory are transformation inputs, not required standalone reporting tables in this implementation.
- Removing staging data destinations avoids unnecessary duplicate Lakehouse tables.
- Re-running against static CSV inputs should not append duplicate customer, product or sales rows.

Step 09-11 | Fix joins and preserve schema

Critical implementation lesson: a merged attribute may be missing from the destination even when present in its source.

GEOGRAPHY ENRICHMENT

Customer	CustomerKey · GeographyKey · Gender
Merge by GeographyKey	Left join matching geography rows
Expand selected fields	City · Country · StateProvince
DimCustomer	Customer attributes with geography

COUNTRY COLUMN RECOVERY

The initial geography expansion omitted **Country**. The Dataflow was edited at the Expand step, Country was included, and the refreshed DimCustomer schema was propagated to the Lakehouse and Direct Lake model.

SCHEMA SAFETY CHECKS

09

Retain stable keys

Keep customer and product identifiers stable so relationships remain valid.

10

Review destination mapping

Check schema mapping and replacement settings before applying new columns.

11

Refresh and confirm

Confirm Country appears in both Lakehouse and semantic-model fields.

Step 12-15 | Build the date dimension

The notebook derives calendar coverage from real orders and inserts only dates missing from the current Delta table.

FINAL LAKEHOUSE TABLES

D1 DimCustomer

Customer attributes, Gender, City, Country

D2 DimProduct

Category, Subcategory, Color, ProductName

D3 FactInternetSales

OrderDate, CustomerKey, ProductKey, sales values

D4 DimDate

Date, Year, Month, Quarter, YearMonth, sort key

UPDATE_DIMDATE NOTEBOOK

12 Attach AdventureWorksLakehouse as default.

13 Read FactInternetSales.OrderDate and find min/max dates.

14 Generate every day across complete earliest/latest calendar years.

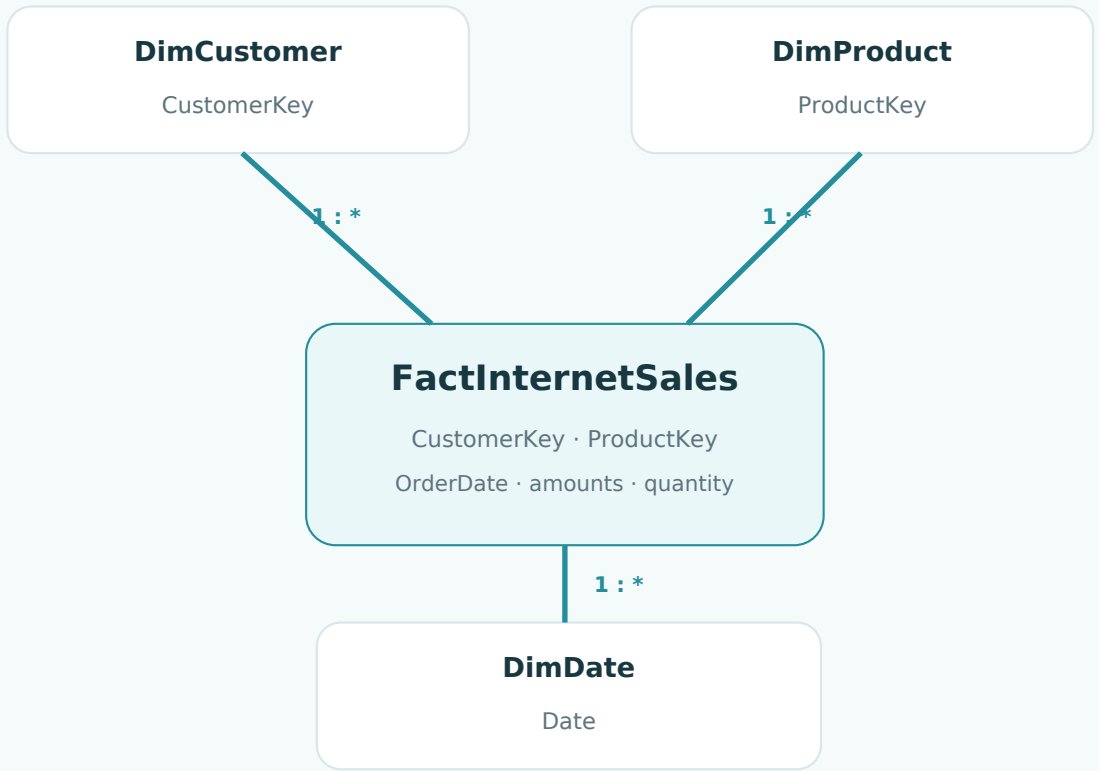
15 Insert missing dates into DimDate and verify coverage.

Safety characteristic: The notebook preserves existing dates and avoids generating sales transactions. A rerun should be idempotent.

06 SEMANTIC MODEL

Step 16-18 | Create Direct Lake relationships

AdventureWorks_Sales_Model reads the Lakehouse Delta tables and provides business-facing DAX calculations.



RELATIONSHIPS

CustomerKey

DimCustomer (1) → FactInternetSales (*)

ProductKey

DimProduct (1) → FactInternetSales (*)

OrderDate

DimDate[Date] (1) → FactInternetSales[OrderDate] (*)

Model checks: matching data types, unique dimension keys, active date relationship, single-direction dimension-to-fact filtering and chronological YearMonthSort.

Implementation approach: star schema with dimension enrichment in Dataflow Gen2.

06 SEMANTIC MODEL

Step 19-21 | Define and verify DAX measures

The report uses central measures rather than re-creating calculations on each visual.

SALES & VOLUME

Total Sales

SUM(FactInternetSales[SalesAmount])

Total Quantity Sold

SUM(FactInternetSales[OrderQuantity])

Sales Records

COUNTROWS(FactInternetSales)

Average Order Value

[Total Sales] / [Total Orders]

CUSTOMER METRICS

Purchasing Customers

DISTINCTCOUNT(FactInternetSales[CustomerKey])

Male Purchasing Customers

CALCULATE by Gender = M

Female Purchasing Customers

CALCULATE by Gender = F

Sales / Purchasing Customer

[Total Sales] / [Purchasing Customers]

COSTS & INSIGHTS

Total Tax

SUM(FactInternetSales[TaxAmt])

Total Freight

SUM(FactInternetSales[Freight])

Best Category

TOPN by Total Sales

Top Sales Month

TOPN by YearMonth

Best Subcategory

TOPN by Total Sales

Critical interpretation: COUNTROWS measures sales records, not necessarily distinct customer orders. Year-over-year comparisons need a valid prior-year period before displaying growth arrows.

07 REPORT DESIGN

Step 22-23 | Establish the visual system

A consistent healthcare-inspired design language was adapted to an actual sales-analysis dataset.

01 Executive Overview

Overall sales KPIs, monthly trend, gender and category

02 Sales Trends

Top sales month, best category, monthly and product trends

03 Product Performance

Top products, color distribution, detailed product table

04 Customer Insights

Purchasing customers, gender, country, city and table

05 Order & Cost Analysis

Quantity, tax, freight, monthly trends and summary

VISUAL STYLE

- Page background
- Cards & visual surfaces
- Primary teal charts
- Secondary teal charts
- Headline text
- Secondary text

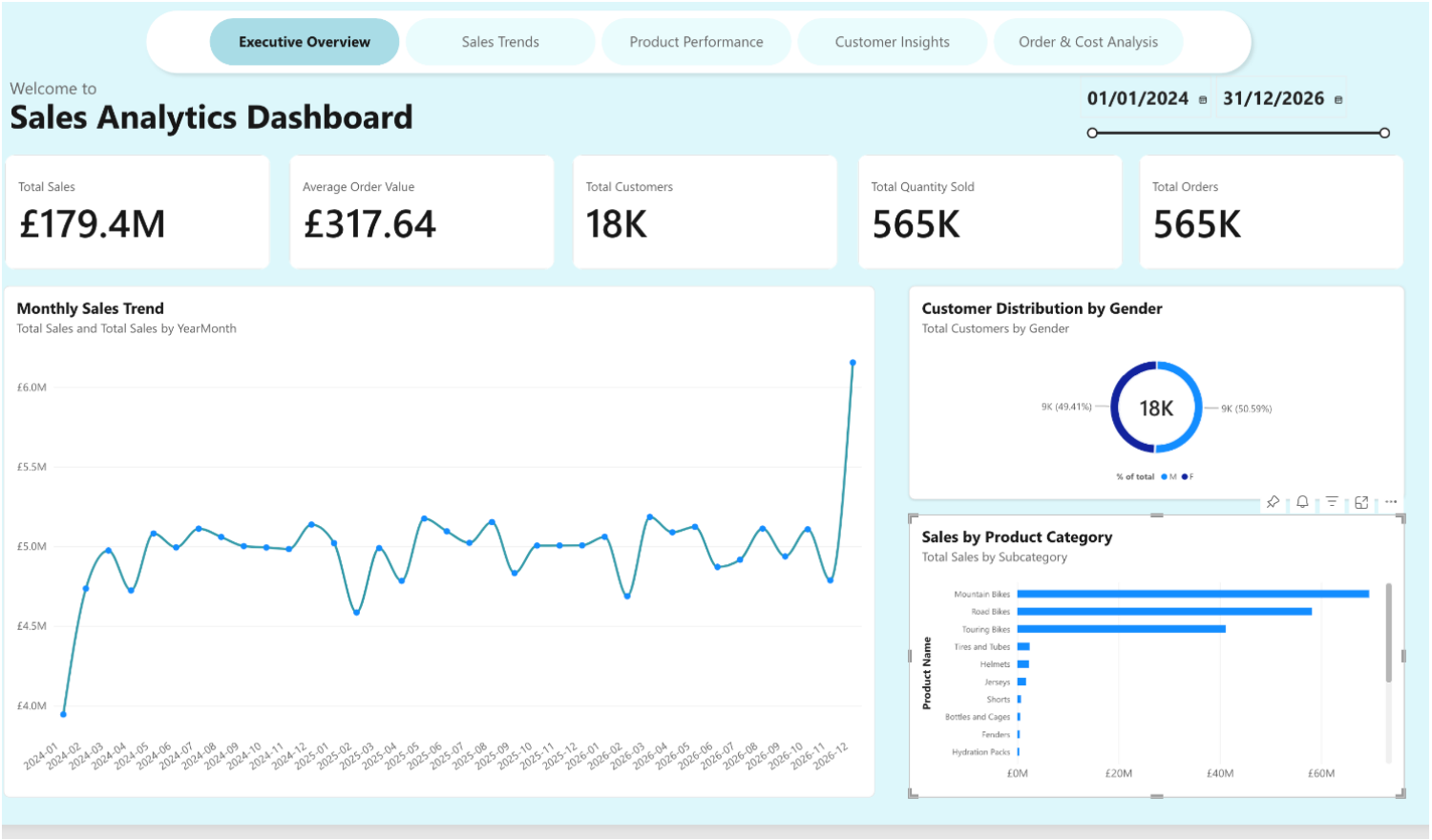
Canvas • 1920 × 1080

Rounded cards • Soft shadows

REPORT SHOWCASE

Executive Overview

The front page combines a single-glance executive scorecard with key sales and customer visuals.



Evidence screenshot from the implemented Executive Overview. Dashboard values are from the user-provided reporting view.

WHAT WAS BUILT

KPI row

Total Sales, Average Order Value, customers, quantity and sales records.

Time series

Monthly sales trend sorted chronologically by YearMonth.

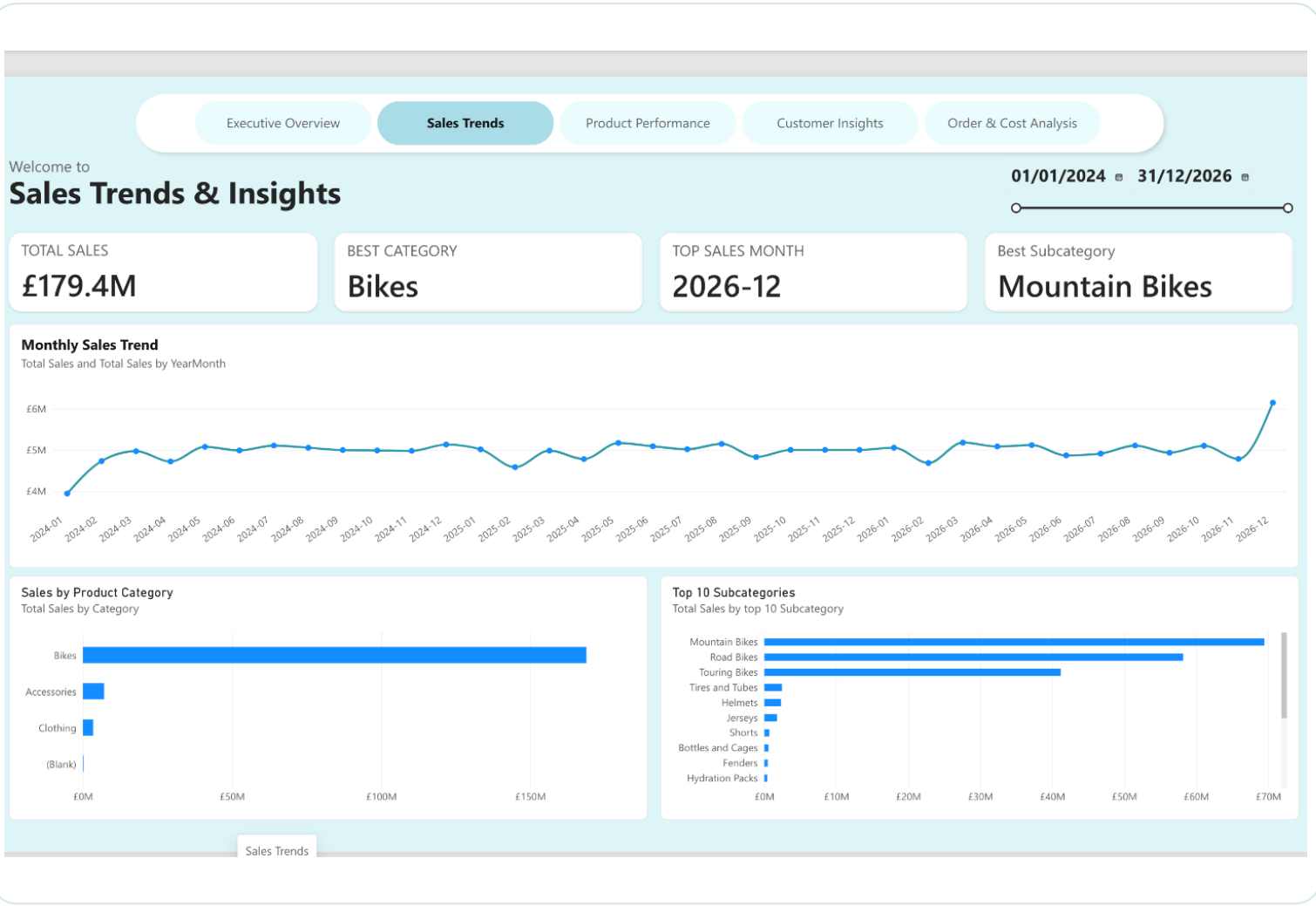
Composition

Gender doughnut and product-hierarchy bar chart complement the trend.

REPORT SHOWCASE

Sales Trends & Insights

Dynamic text measures identify the best-performing category, month and subcategory.



Captured view includes Best Category = Bikes, Top Sales Month = 2026-12 and Best Subcategory = Mountain Bikes.

WHAT WAS BUILT

Insight cards

Total Sales, Best Category, Top Sales Month and Best Subcategory.

Trend analysis

Monthly sales line with category and Top 10 subcategory charts.

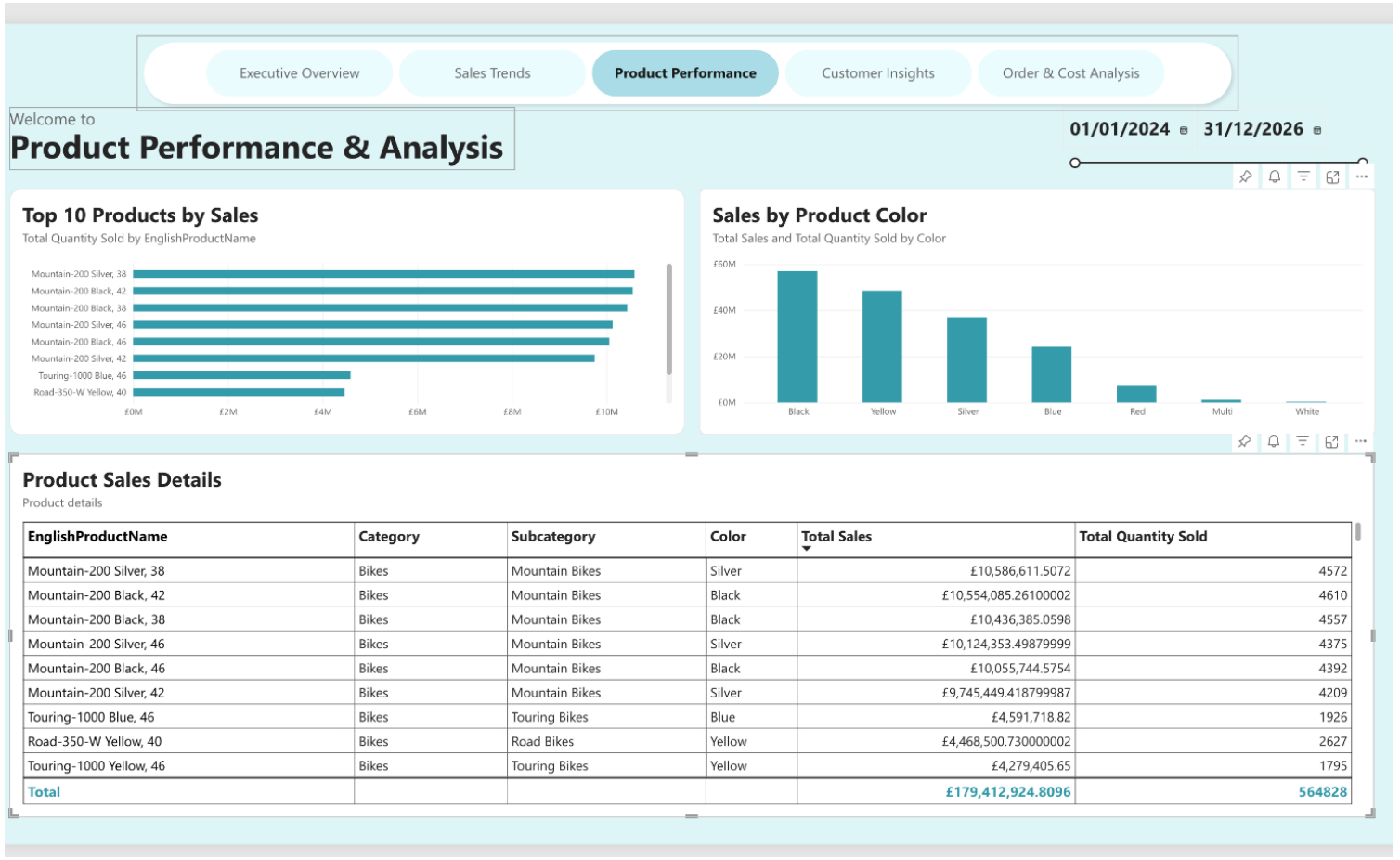
Filter-aware

DAX insights respond to the selected Date slicer context.

REPORT SHOWCASE

Product Performance

Product-level comparisons and category drill-down support merchandising analysis.



EnglishProductName	Category	Subcategory	Color	Total Sales	Total Quantity Sold
Mountain-200 Silver, 38	Bikes	Mountain Bikes	Silver	£10,586,611.5072	4572
Mountain-200 Black, 42	Bikes	Mountain Bikes	Black	£10,554,085.26100002	4610
Mountain-200 Black, 38	Bikes	Mountain Bikes	Black	£10,436,385.0598	4557
Mountain-200 Silver, 46	Bikes	Mountain Bikes	Silver	£10,124,353.49879999	4375
Mountain-200 Black, 46	Bikes	Mountain Bikes	Black	£10,055,744.5754	4392
Mountain-200 Silver, 42	Bikes	Mountain Bikes	Silver	£9,745,449.418799987	4209
Touring-1000 Blue, 46	Bikes	Touring Bikes	Blue	£4,591,718.82	1926
Road-350-W Yellow, 40	Bikes	Road Bikes	Yellow	£4,468,500.730000002	2627
Touring-1000 Yellow, 46	Bikes	Touring Bikes	Yellow	£4,279,405.65	1795
Total				£179,412,924.8096	564828

Actual Product Performance & Analysis report screenshot; table demonstrates product-level drill-down.

WHAT WAS BUILT

Top 10 products

Ranked horizontal bars using EnglishProductName and Total Sales.

Product colors

Column chart comparing sales by the Color attribute.

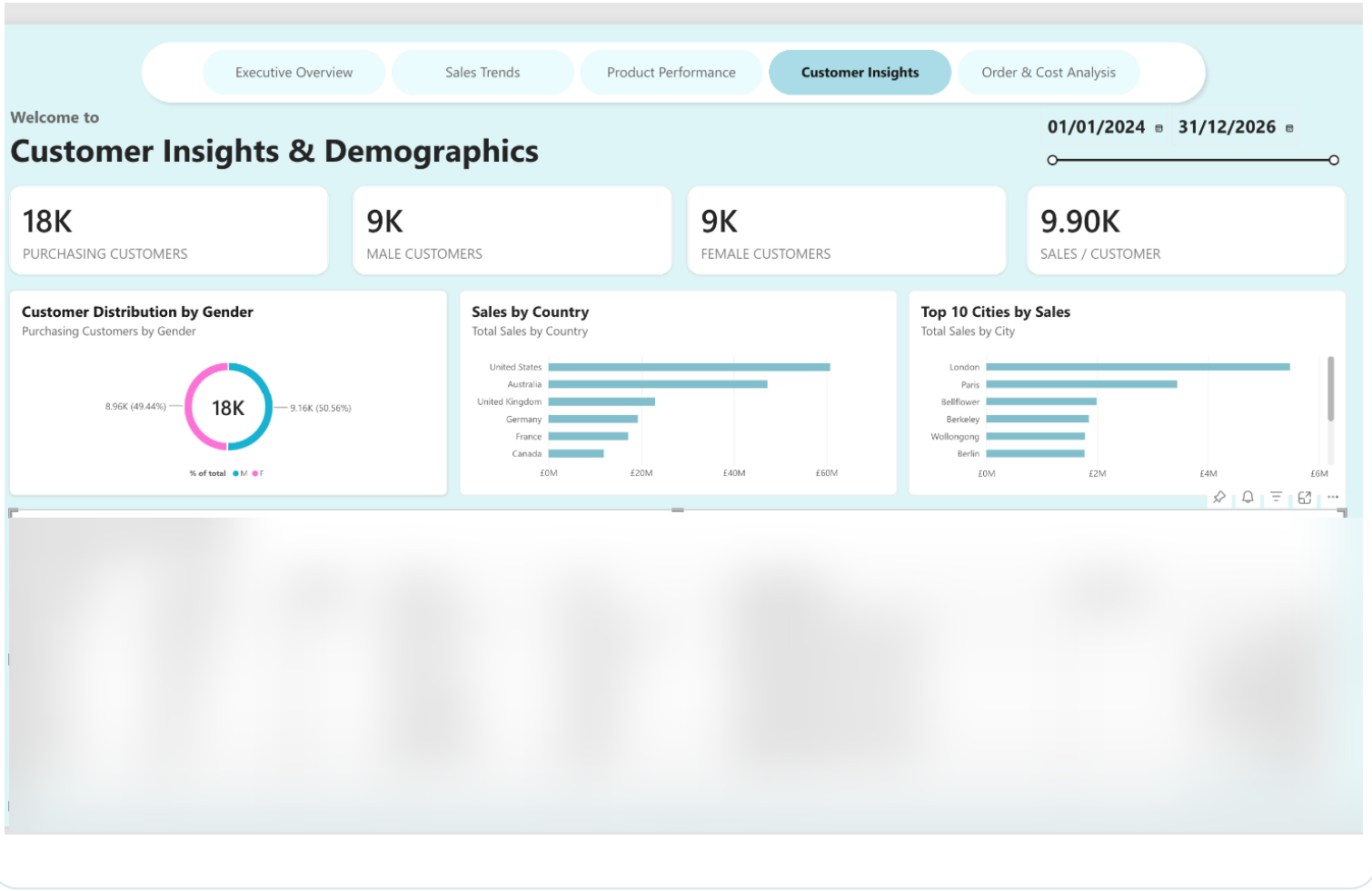
Detail table

Product, Category, Subcategory, Color, Sales and Quantity.

REPORT SHOWCASE

Customer Insights

The enriched customer dimension supports demographics and geographic breakdowns.



Customer Insights & Demographics screenshot. Customer detail rows are intentionally obscured for safe distribution.

WHAT WAS BUILT

Customer metrics

Purchasing customers, male/female purchasers and sales per customer.

Geographic view

Sales by Country and Top 10 Cities by Sales.

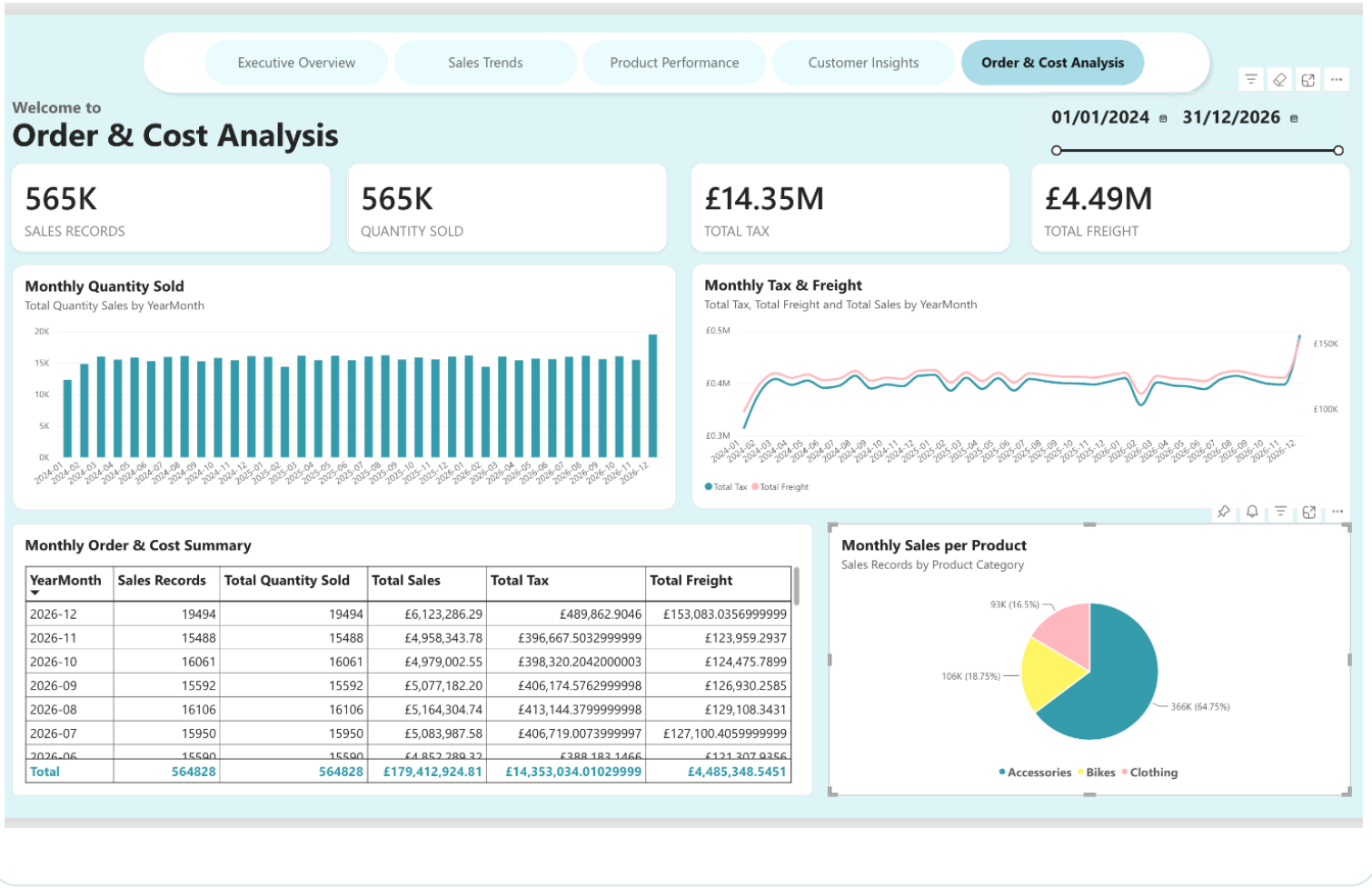
Privacy care

Row-level customer identifiers and emails are blurred in this document.

REPORT SHOWCASE

Order & Cost Analysis

The final report page tracks volume, tax, freight and month-by-month activity.



YearMonth	Sales Records	Total Quantity Sold	Total Sales	Total Tax	Total Freight
2026-12	19494	19494	£6,123,286.29	£489,862.9046	£153,083.0356999999
2026-11	15488	15488	£4,958,343.78	£396,667.5032999999	£123,959.2937
2026-10	16061	16061	£4,979,002.55	£398,320.2042000003	£124,475.7899
2026-09	15592	15592	£5,077,182.20	£406,174.5762999998	£126,930.2585
2026-08	16106	16106	£5,164,304.74	£413,144.3799999998	£129,108.3431
2026-07	15950	15950	£5,083,987.58	£406,719.0073999997	£127,100.4059999999
2026-06	15590	15590	£4,852,289.32	£388,183.1466	£121,307.9356
Total	564828	564828	£179,412,924.81	£14,353,034.01029999	£4,485,348.5451

Order & Cost Analysis screenshot from the completed report; recorded values include £14.35M tax and £4.49M freight.

WHAT WAS BUILT

Cost KPIs

Sales records, quantity sold, Total Tax and Total Freight.

Monthly comparisons

Quantity Sold bars and Tax/Freight lines over YearMonth.

Cost summary

Monthly detail table and category-based sales-record distribution.

08 KPI INTELLIGENCE

Step 24 | Add up / down comparisons

Trend mini-cards make the scorecards actionable when a meaningful comparison period is available.

TOTAL SALES

£179.4M

▲ +x.xx% vs prior year

QUANTITY SOLD

565K

▼ −x.xx% vs prior year

TOTAL TAX

£14.35M

— if prior year missing

MEASURE DESIGN

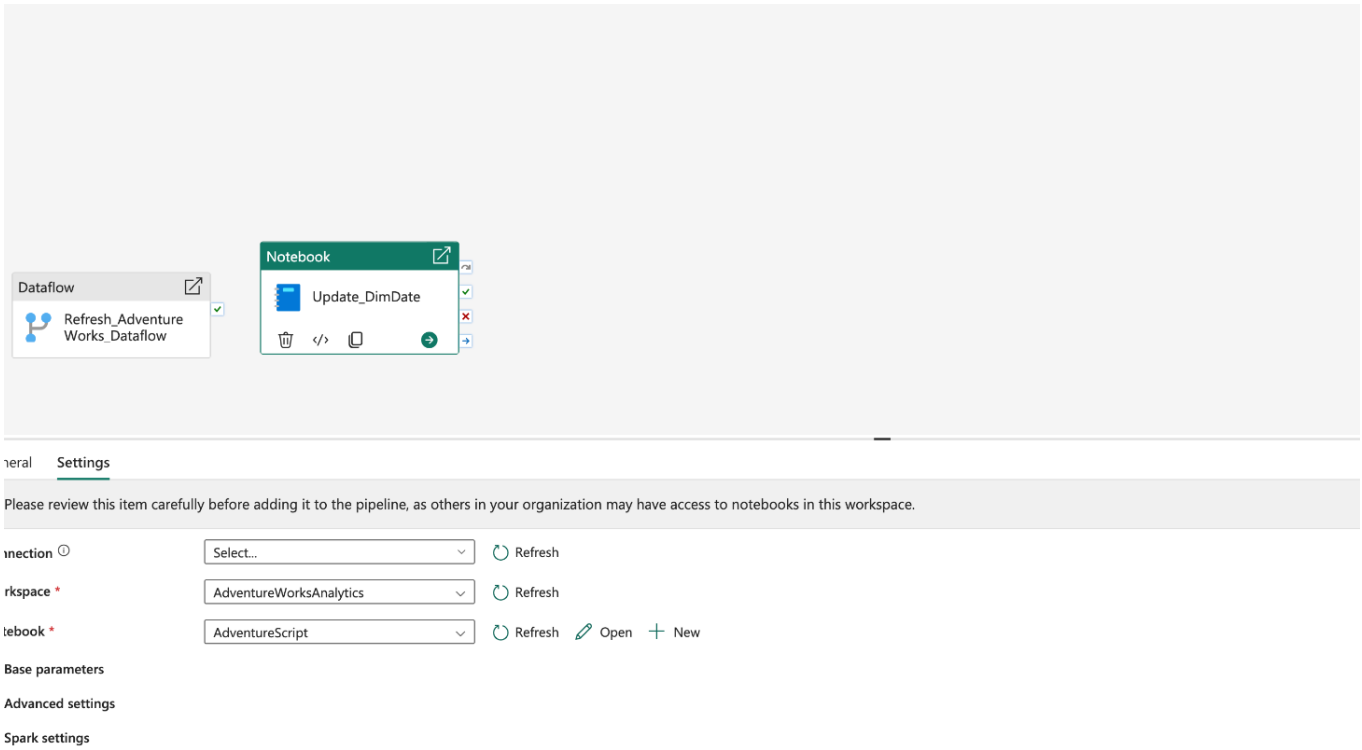
01	**Previous period**	CALCULATE([Total Sales], SAMEPERIODLASTYEAR(DimDate[Date]))
02	**Percentage change**	DIVIDE([Total Sales] - [Total Sales PY], [Total Sales PY])
03	**Display & color**	FORMAT(value,"0.00%") + ▲/▼ and a conditional hex color

Quality rule: A missing or zero prior-year value should show a neutral/no-comparison state. A tax/freight increase is not inherently favorable, so use neutral or contextual colors for cost KPIs.

09 AUTOMATION

Step 25-27 | Orchestrate the ETL pipeline

The Dataflow activity updates Lakehouse tables; the Notebook activity maintains the calendar after success.



PIPELINE ITEMS

25 Create PL_AdventureWorks_Daily_ETL in the workspace.

26 Add Dataflow activity referencing DF_AdventureWorks_Transformation.

27 Add Update_DimDate Notebook after green Success connection.

Important: This screenshot shows notebook configuration during setup, not proof of a finished scheduled run.

Notebook configuration screenshot captured during pipeline setup. Activities and connections may vary slightly by Fabric workspace interface.

09 AUTOMATION

Step 28-30 | Test, synchronize, schedule

Complete pipeline operations should be validated against the Lakehouse and finished report before unattended runs.

28 Validate and run

Pipeline editor → Validate → Save → Run. In run output, inspect Dataflow and Notebook activity status and failure details.

29 Synchronize Direct Lake

Verify the semantic model setting Keep your Direct Lake data up to date. Add explicit model refresh only if supported and required.

30 Schedule and monitor

Use Pipeline → Schedule to set recurrence and time zone. Use Monitoring hub / View run history to review each execution.

Evidence boundary: The date notebook was confirmed working. This document does not assume that a recurring schedule or the optional semantic-model-refresh activity has been enabled.

Acceptance checklist | Data, model, visuals and automation

Use these checks after the first manual run and periodically after source or schema changes.

CONTROL	EXPECTED VERIFICATION
CSV source	Expected file names, columns and permitted source access
Dataflow Gen2	Successful refresh, valid joins and intended destination behavior
Lakehouse	No duplicate append; required Delta tables and rows present
DimDate	Every OrderDate in calendar; dates sort chronologically
Semantic model	Active relationships; measures give expected filter results
Executive page	KPI values and category labels are consistent
Sales/Product pages	Top N, Month and product drill-downs respond to filters
Customer page	Country/City attributes and purchasing counts are valid
Order/Cost page	Tax/Freight decimals, quantity measure and chart legend verified
Automation	Run history succeeds; schedule enabled only after successful manual test

Baseline visible in the report: Total Sales ≈ £179.4M, sales records ≈ 564,828, purchasing customers ≈ 18K, Total Tax ≈ £14.35M, Total Freight ≈ £4.49M. These reflect the captured data and filter context, not guaranteed future values.

11 HANDOVER

Operational notes and enhancement roadmap

A maintainable report depends on trusted inputs, clearly named metrics, and observable pipeline executions.

What is completed

- CSV ingestion, Dataflow Gen2 transformations and three loaded analytical tables.
- DimDate notebook and four-table Direct Lake model.
- Five report pages, interactive measures and KPI indicators.
- Pipeline created and Notebook confirmed functional.

Operational items to confirm

- Pipeline manual run succeeds end to end after final connector configuration.
- Recurring schedule and time zone are enabled as intended.
- Direct Lake update setting or explicit model sync works after ETL.
- Date records through December 2026 are intentional if used as demo data.

RECOMMENDED NEXT ITERATION

- Introduce a stable SalesOrderNumber if true distinct-order counting is needed.
- Add pipeline failure alerts and source file freshness checks.
- Promote repeated card formats into a centrally managed theme and report template.
- Add role-based access / data privacy checks before wider distribution.

APPENDIX

Implementation sequence & item reference

Compact index of the actual Fabric assets and the major decisions made in the build.

COMPONENT	IMPLEMENTED ARTIFACT OR DECISION
Source CSVs	Customer, geography, Product, ProductCategory, ProductSubcategory, internet_sales
Dataflow Gen2	DF_AdventureWorks_Transformation
Lakehouse	AdventureWorksLakehouse
Delta tables	DimCustomer · DimProduct · FactInternetSales · DimDate
Notebook	Update_DimDate_AdventureWorks / Update_DimDate
Semantic model	AdventureWorks_Sales_Model (Direct Lake)
Power BI report	Sales Analytics Dashboard (five pages)
Pipeline	PL_AdventureWorks_Daily_ETL
Orchestration	Dataflow refresh → Success → Notebook → Direct Lake synchronization
Validation	Refresh history, table checks, slicer interactions, financial formatting

Documentation basis: the build conversation, actual report screenshots, attached notebook and Draw.io/DFD assets.